

计算机网络练习题(简答&综合专题)

IV. Short Answer Questions (20 Points)

1. List the 5 layers of the Internet Protocol Stack. Why do we layer protocols? (3 points)
2. Draw the 3-way handshake diagram. Why is a 3-way handshake used instead of a 2-way handshake? (4 points)
3. Briefly explain one advantage of LS over DV and one advantage of DV over LS. (3 points)
4. Explain how a switch populates its forwarding table (Switch Table). What happens if a destination MAC is not in the table? (4 points)
5. What is the difference between a MAC address and an IP address in terms of their role in packet delivery (i.e., when is each used)? (3 points)
6. Name the two phases of TCP Congestion Control (Reno) shown in a standard CWND vs Time graph. What event typically causes the transition from the first phase to the second? (3 points)

V. Comprehensive Problems (30 Points)

Problem 1: IP Addressing & Subnetting (10 Points)

An ISP is granted a block of addresses starting with 192.168.100.0/24. The ISP needs to distribute these addresses to three organizations (Org A, Org B, Org C) with the following requirements:

- Org A needs 60 addresses.
- Org B needs 120 addresses.
- Org C needs 30 addresses.

A. Assign subnet addresses (CIDR notation) to Org B first, then Org A, then Org C to satisfy the needs without overlapping. (6 points)

- Org B Subnet: _____ (Range: _____ to _____)
Org A Subnet: _____ (Range: _____ to _____)
Org C Subnet: _____ (Range: _____ to _____)

B. What is the subnet mask (in dotted decimal) for Org A? (2 points)

C. How many addresses are left unassigned in the original /24 block? (2 points)

Problem 2: Reliable Data Transfer (10 Points)

Consider the Stop-and-Wait protocol.

Host A sends packets of size $L=1000$ Bytes to Host B over a link with distance $d=2500$ km, propagation speed $v=2.5 \times 10^8$ m/s, and transmission rate $R=1$ Mbps.

A. Calculate the Propagation Delay (d_{prop}) and Transmission Delay (d_{trans}). (4 points)

B. Assuming no packets are lost and the ACK size is negligible (0 transmission time), calculate the total time from when A starts sending a packet until it receives the ACK. (3 points)

C. Calculate the Utilization (U_{sender}) of the channel. Is Stop-and-Wait efficient in this scenario? Why or why not? (3 points)

Problem 3: Link State Routing (Dijkstra) (10 Points)

Consider the network shown below with nodes A, B, C, D, E.

Link Costs:

A-B: 2

A-D: 1

B-C: 3

B-D: 2

D-C: 5

D-E: 1

C-E: 1

Task: Use Dijkstra's algorithm to calculate the shortest path from Node A to all other nodes. Show the steps in a table.

Format: (Cost, Predecessor). e.g., (2, A) means cost is 2, previous hop is A.

简答题答案:

1. 层次:应用层、传输层、网络层、链路层、物理层。原因:将复杂系统解耦, 模块化, 便于更新维护, 某层变化不影响其他层。
2. 握手:SYN -> SYN/ACK -> ACK。原因:防止已失效的历史连接请求突然传到服务端产生错误连接;同步双方初始序列号。
3. 比较:LS 优点是收敛快、不易产生环路;DV 优点是算法简单、控制消息流量较小(仅与邻居交换)。
4. 交换机:通过检查进入帧的源 MAC 地址来学习, 记录该 MAC 对应的接口。如果目的 MAC 不在表中, 则向除入接口外的所有接口泛洪 (Flood)。
5. 区别:IP 地址用于端到端(跨越不同网络)的路由和寻址;MAC 地址用于逐跳(同一局域网/链路内)的帧传输。
6. 阶段:1. 慢启动 (Slow Start);2. 拥塞避免 (Congestion Avoidance)。事件:当拥塞窗口 (cwnd) 达到慢启动阈值 (ssthresh) 时。

综合题答案:

1.

A.分配:

组织 B (120):需要 128 个地址 (/25)。分配 192.168.100.0/25 (范围 .0 - .127)。

组织 A (60):需要 64 个地址 (/26)。紧接着 B 分配, 192.168.100.128/26 (范围 .128 - .191)。

组织 C (30):需要 32 个地址 (/27)。紧接着 A 分配, 192.168.100.192/27 (范围 .192 - .223)。

B.子网掩码 (A):/26 对应 255.255.255.192。

C.剩余:范围从 .224 到 .255。共 $256-128-64-32=32$ 个地址。

2.

A.时延计算:

$d_{\text{prop}}=2500 \times 10^3 / 2.5 \times 10^8=0.01\text{s}=10\text{ms}$ 。

$d_{\text{trans}}=1000 \times 8 / 1 \times 10^6=0.008\text{s}=8\text{ms}$ 。

B.总时间:

$T=d_{\text{trans}}+d_{\text{prop}}(\text{去})+d_{\text{prop}}(\text{回, ACK})=8+10+10=28\text{ms}$ 。(注:题目假设ACK传输时间忽略, 但传播时间不能忽略)。

C.利用率:

$U= d_{\text{trans}}/T \approx 0.286$ (或 28.6%)。

结论:效率不高。因为大部分时间(20ms)都在等待 ACK, 链路处于空闲状态。

3.

Step	N'	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)
0	A	2, A	$\infty \infty$	1, A	$\infty \infty$
1	A, D	2, A	6, D	1, A	2, D
2	A, D, B	2, A	5, B	1, A	2, D
3	A, D, B, E	2, A	3, E	1, A	2, D
4	A, D, B, E, C	2, A	3, E	1, A	2, D

(注: Step 2 选择 E 或 B 均可, 顺序可能略有不同, 但最终结果一致)

A的转发表:

目的地 (Destination)	下一跳 (Next Hop)
B	B
C	D
D	D
E	D